Examiner reports

Q1.

This question proved to be a good source of marks for many candidates. In part (a)(i) most candidates quoted the correct value for the power p. Although many candidates produced correct answers for the integration, all the usual errors, as illustrated by $\frac{3}{2}x^{1.5}$, $x^{1.5}$, $\frac{x^{-0.5}}{-0.5}$, and $\frac{\sqrt{x}^2}{2}$, were seen. Most candidates stated and used the correct limits in part (a)(iii) but evaluation was sometimes poor. Many of the candidates found a correct equation for the tangent but it was surprising to find a significant minority of these candidates either making no attempt to find the area of triangle AOB or giving the area as a negative value. For full marks in part (c) the examiners were expecting to see the word 'translation'. Some candidates wrote 'tr' but this was not considered to be clear enough to distinguish between 'transformation' and 'translation'. Common wrong answers involved 'stretch' and vectors in the wrong direction. In part (d) there was a significant minority of candidates starting their solution with the incorrect statement $\sqrt{x-1} = \sqrt{x} - \sqrt{1} = \sqrt{x} - 1$ but in general the trapezium rule was well understood and many candidates were able to give the final answer to the correct required degree of accuracy in this numerical integration question.

Q2.

Part (a) was well answered, although some candidates integrated. There were many fully correct answers to part (b), but a lot of candidates chose the incorrect functions to equal u and $\mathrm{d}v/\mathrm{d}x$ with the obvious ensuing problem.

Part (c) was poorly answered. Candidates must realise that, in any substitution question, apart from replacing functions of x with functions of u, they must also find $\mathrm{d}u/\mathrm{d}x$. A large number of candidates attempted to integrate an expression containing both x's and u's.

Q3.

This question discriminated well between candidates. Part (a) showed an alarming lack of the correct use of mathematical language. Candidates had to use the words translation and stretch. Centres must stress this to all their students. There were many correct solutions to part (b) although some candidates missed the final mark by giving their answer as a decimal rather than the exact value that was required. Many candidates scored some marks on the final part but there were few fully correct solutions.

Q4.

Part (a) This was done very well with most candidates showing confidence and accuracy with the partial fractions. Both techniques of substituting suitable values of x or determining unknown coefficients using simultaneous equations were seen. Any errors made were usually arithmetical.

Part (b) Most candidates knew that they were to use the partial fractions in attempting the integral, and most knew these were 'ln' integrals, although some attempts at independently integrating numerator and denominator were seen. The common error in the ln integrals was the omission of the coefficient 1/2.

Q5.

Part (a) Most candidates successfully wrote down the expression for sin $2x$.

Part (b)(i) This caused difficulties for some candidates. Fairly common answers were $2\cos 2x = 2\cos^2 x - 2\sin^2 x$ or $\cos 2x = \cos^2 x + \sin x$.

Part (b)(ii) Most candidates made a reasonable start at proving the identity with many more able to handle the given $x + 2x$ than could consider $x + x$ for themselves. Candidates' responses varied considerably from clear concise derivations, to dead ends and restarts, to long meandering routes; some expanding cos$2x$ and then bringing it back in. Most candidates knew they had to eliminate sin x at some point, but this was not always done convincingly.

Part (c) Those candidates who saw the connection to part (b)(ii) were usually successful in their integration, although there were some sign errors and also errors in handling the 4. Some candidates wrote $\cos x^3$ as $(1 - \sin^2 x)\cos x$ and went on to integrate correctly. Attempts by parts, and even by substitution, went awry. Many candidates opted for the somewhat hopeful $\sin^4 x$, or similar, as their integral but could get some credit for correct use of the given limits.

Q6.

This question was also answered well by the vast majority of students, who showed that they knew the techniques involved in this question. In part (a) the majority of students integrated to find the velocity of the particle but a common error was to forget the + $\mathbf{c}$ term and thus to ignore the initial position vector. Some found the + $\mathbf{c}$ incorrectly, assuming that $\mathbf{c}$ was $6\mathbf{i} - 5e^{-4}\,\mathbf{j}$ without completing the necessary calculation.
Another common error was seen whereby students did not integrate the e^{-4t} term correctly. In part (b) a number of students found the initial velocity, $-15\mathbf{i} - 5\mathbf{j}$, but did not find the initial speed.

Q7.

Most students answered part (a) correctly but some did not replace their a by $\frac{\mathrm{d}v}{\mathrm{d}t}$. In part (b) students were expected to use $\int \frac{1}{100-v}\,\mathrm{d}v = -\int \frac{1}{40}\,\mathrm{d}t$; some incorrectly split $\int \frac{1}{100-v}\,\mathrm{d}v$ into $\int \frac{1}{100} - \frac{1}{v}\,\mathrm{d}v$, whilst others incorrectly used $\int (100-v)\,\mathrm{d}v = \int \frac{1}{40}\,\mathrm{d}t$.

A common error was in forgetting the minus sign in the evaluation of $\int \frac{1}{100-v}\,\mathrm{d}v$. Most students attempted to find the '+ c' term in the solution of the differential equation.

Q8.

This question was also answered well. In part (a), a few candidates did not divide **F** by 50 accurately. In part (b), a number of candidates did not find their '+$\mathbf{c}$' correctly, simply replacing their '$\mathbf{c}$' with $7\mathbf{i} - 4\mathbf{j}$. Others, on evaluating their '$\mathbf{c}$', took $-e^{-2t}$ when $t = 0$ to be +1 rather than –1.

Q9.

This question was answered well by most candidates. However, in part (a), a few candidates did not show the required step 0.4 $\frac{dv}{dt}$ = 2 – 4v ; others lost and recreated the minus signs, eg $\frac{dv}{dt}$ = 5 – 10v = – 10 (0.5 + v) = – 10 (v – 0.5), which was penalised. In part (b), the majority of candidates used $\int \frac{1}{v-0.5} dv = -\int 10dt$ and then solved this by integrating both sides.

The integration $\int \frac{1}{v-0.5} dv$ caused some candidates difficulty, finding terms such as $\frac{1}{\frac{1}{2}v^2 - 0.5v}$ etc. The use of logarithms was often incorrect, with ln (v – 0.5) = –10t + ln 0.5 commonly becoming v – 0.5 = e – 10t + 0.5

Q10.

This question was also answered well by virtually all candidates, who showed that they knew the techniques involved in answering it. In part (c) the majority of students integrated to find the position vector but a common error was to forget the + c term and thus to ignore the initial position vector. Some found the + c incorrectly assuming that **c** was zero without completing the necessary calculation.

Q11.

Most candidates answered part (a)(i) correctly but some just wrote down the printed result.
In part (a)(ii), candidates needed to start from $F = ma$ and show the cancellation of the mass (65). Some were penalised for ignoring the m term or ignoring the minus sign.

In part (b), candidates were asked to show that $\int \frac{1}{v-2.45} dv = -\int 4dt$ many simply used this equation and were penalised. There were, however, many good answers to this part, with both the integrations required being successfully completed. The difficulty which candidates had was in the simplification needed to convert ln(v – 2.45) = – 4t + ln17.15 into
v – 2.45 = 17.15e^{-4t} , which often became v – 2.45 = 17.15 + e^{-4t}.

Q12.

Most candidates showed that they knew the techniques involved in answering this question. Candidates generally answered part (a) correctly. Part (b)(i) was answered well, but in part (b)(ii) some candidates forgot to find the magnitude of **F**, giving instead just the vector **F**, – 40**i** + 30**j**. Most candidates answered part (c) correctly. In part (d) the majority of candidates integrated to find the position vector, but a common error was to forget the +**c** term and thus to ignore the initial position vector. Some found the + **c** incorrectly, assuming that **c** was the initial position vector.

Q13.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good answers to part (b), although a significant number of candidates used 'inventive' algebra. A few candidates who integrated to find the equation in v and t forgot the $+c$ term.

Q14.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good answers to part (b), although some used inventive algebra. The solution to part (c) was usually correct, although candidates often forgot to obtain ± 1, for the square root of v, when they found the square-root of the equation obtained in part (b).

In part (d) candidates who integrated v to find the distance travelled often forgot the + d term which was required to be shown to be equal to zero.

Those who used integration with limits commonly used $\int_{16}^{1}(4-0.1t)^2\,dt$ forgetting that the integral was with respect to t so that $\int_{0}^{30}(4-0.1t)^2\,dt$ was required. A considerable number of candidates, in part (d), tried to use equations for motion with constant acceleration.

Q15.

The majority of candidates used $\frac{dv}{dt} = -\frac{\lambda}{v^{\frac{1}{4}}}$ and then solved this by using $\int v^{\frac{1}{4}}\,dv = \int -\lambda\,dt$. Some used $a = \frac{dv}{dt}$, together with $v = u + at$, so that $v = u - \frac{\lambda}{v^{\frac{1}{4}}}t$ was often seen.

Many candidates found $\frac{4}{5}v^{\frac{5}{4}} = -\lambda t + \frac{4}{5}y^{\frac{5}{4}}$, but the solution for v was not often found correctly.

A common result was "$v^{\frac{5}{4}} = u^{\frac{5}{4}} - \frac{5}{4}\lambda t \Rightarrow v = u - \left(\frac{5}{4}\lambda t\right)^{\frac{4}{5}}$".

Q16.

This question was usually answered correctly, although some obtained –4cos2t rather than +4cos2t. A common error was the omission of $+c$, while others added $+c$ but automatically assumed that c = 0.

Q17.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for starting with $\frac{dv}{dt} = -\frac{0.08v^2}{0.05}$. There were many good answers to part (b), with very few candidates being unable to obtain $v = \frac{15}{5 + 24t}$ from $\frac{1}{v} = 1.6t + \frac{1}{3}$.

Q18.

Part (a) was usually answered correctly, although some obtained $+6\cos 3t$, rather than $-6\cos 3t$. In part (b), the common error was the omission of $+c$, while others added $+c$ and automatically assumed that $c = 0$.

Q19.

Parts (a), (b) and (c) were answered well. In part (d), few candidates used the fact that the work done by the force was equal to the change in kinetic energy. Most used complicated methods which rarely gave an appropriate answer.

Q20.

Virtually all candidates obtained $\frac{dv}{dt} = -0.05v$: only a few ignored the required step $m\frac{dv}{dt} = -0.05mv$. The equation $\int \frac{dv}{v} = -\int 0.05 \, dt$ was a necessary step which needed to be seen in part (b). Candidates knew roughly how to obtain $v = 20e^{-0.05t}$. Unfortunately, too often algebraic skills were not sufficient and the equation
$\ln v = -0.05 + c$ regularly became $v = e^{-0.05t} + c$ before becoming $v = Ae^{-0.05t}$. This and similar errors were not condoned.

Q21.

Most candidates completed this question successfully. A few used $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ in part (b) and obtained $\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 6t^2)\,\mathbf{j}$, rather than the given answer. After they had done the integration of **a**, a number of candidates 'invented' the 6**i** + 30**j** terms required.

Q22.

A significant number of candidates created $\frac{dv}{dt} = -\lambda v$ without proper justification. The equation $-\lambda mv = ma$ was a necessary step which needed to be seen. Even more knew roughly how to obtain $v = Ue^{-\lambda t}$, but too often algebraic skills were not sufficient and
$\ln v = -\lambda t + \ln u$ regularly became $v = e^{-\lambda t} + U$ before becoming $v = Ue^{-\lambda t}$. This and similar errors were not condoned.

Q23.

Quite a few candidates had problems dealing with the differentiation and integration of the exponential function in this question. They used techniques that were confused and were unable to obtain the correct results. This was probably due to the fact that this question demanded the use of techniques from the core mathematics modules, with which the candidates were not yet fully confident. In part (a)(ii), very few candidates gave fully correct answers for the range of values of the acceleration. Some obtained either the value 2 or the value 14. Of those that did obtain both values, very few used the correct inequalities. In part (b), there were problems with the integration of the exponential function and also with candidates omitting or not find the constant of integration.

Q24.

The responses to the modelling in parts (a) and (b) were very varied. Only a small number of candidates gained all three marks, but most gained at least one mark. Some candidates made statements that contradicted the information in the stem of the question, such as “the resistance force will be constant”. Part (c) tended to be done very well or very badly. There was some use of constant acceleration equations, but this was not very common. The biggest issue was the omission of a negative sign in the formation of the differential equation, which then continued throughout the solution. This error was quite common. Other problems were caused by incorrect algebraic manipulation, particularly with logarithms, or integration.

Q25.

Part (a) of this question was often done well. Many candidates seemed to take the question in their stride, but others made limited progress or did not know how to approach the question. There were some who lost accuracy marks. The four main causes of this were;

- Omission of the negative sign in the differential equation,
- Algebraic errors,
- Incorrect integration,
- Not finding the value of the constant of integration.

Part (b) was done very well, with many candidates gaining the available mark without doing part (a).

Q26.

The candidates generally found part (a) of this question very difficult. Some of the more able candidates found an expression for F in terms of t and then applied Newton’s second law. A few effectively reversed this process, by changing the graph into one for acceleration and then finding the equation of the line. Due to the fact that the answer was given, there were a large number of partial or very unconvincing arguments. Some candidates simply omitted this part of the question and went straight on to part (b).

There were many good responses to part (b), but a common error was to ignore the constant of integration. Candidates do need to include a constant of integration and find its value, even if it is zero, in order to gain full marks.

There was a similar problem in part (c), with many candidates again ignoring the constant of integration. Those candidates who included limits of integration, experienced less difficulty, because they wrote down the values of 0 and 200 and substituted both of them.

A few candidates used constant acceleration equations, but the number doing this was fairly small.

Part (d) produced some very mixed responses. There were some very good explanations, but some were very confused. Some candidates wrote about the acceleration instead of the velocity. In some cases the candidates seemed to have some idea of how to respond, but did not express this clearly. Quite a few candidates said that the constant would change, which seemed to imply that they were thinking about the intercept of the graph, but this did not lead to any marks as the velocity does not contain a constant term.

Q27.

Many candidates produced fully correct and succinct solutions here. Others barely scored any marks, as they seemed unaware of the variable nature of the acceleration and applied the equations of constant acceleration. A few started correctly, but made errors in separating variables prior to integration.

Q29.

This question was generally done well, but there was one fairly common error that was seen on quite a few scripts. The same problem arose in both parts (a) and (c), where the candidates either did not introduce a constant of integration or did not show that it was zero. There were also a few minor errors that were produced when integrating, but these were fairly rare. A few candidates tried to use constant acceleration equations, but this approach was also quite unusual.

Q30.

Some candidates did this very well. Almost all of them realised that integration was required, but some were unable to carry this out. There were a variety of errors, such as omitting the constant of integration and multiplying by 4 instead of dividing, but the most serious was to increase the power of t to give results such as 5 cos t^2. Errors made in part (a) were often repeated in part (b).

Q31.

Part (a) of this question was done well by almost all candidates. There were a lot of good solutions for the rest of the question, but there were two problems that emerged. The first was concerning the constants of integration. Some candidates did not include these in their working and simply added a 6 to the velocity component in part (b) to get the right answer. Candidates do need to show that they introduce constants when integrating and that they do evaluate them. The other problem emerged less frequently, but some candidates did try to use constant acceleration equations to find the velocity and position vectors.

Q32.

This question was generally done very well. The vast majority of candidates gained full marks on part (a). In part (b), some candidates did not take account of the initial conditions, although most of these candidates did integrate correctly. A few candidates did make minor errors when trying to find the constant of integration.